ANSWER SCHEME
CHAPTER 3: ELECTRIC CURRENT AND DIRECT CURRENT CIRCUITS

No.	Answer Scheme
1	(a) $V = IR$ $12 = I(6)$ $I = 2$ A (b) $I = \frac{Q}{t}$ $2 = \frac{Q}{2}$ $Q = 4$ C
2	(a) $I = \frac{Q}{t}$ $I = \frac{Ne}{t}$ $5 = \frac{N(1.60 \times 10^{-19})}{5}$ N = 1.56×10^{20} electrons (b) $Q = It$ $= (5)(10)$ $= 50$ C
3	$R = \frac{\rho l}{A}$ $= \frac{(2.82 \times 10^{-8})(15000)}{5.0 \times 10^{-4}}$ $= 0.846\ \Omega$
4	$R = \frac{V}{I}$ $= \frac{1.5}{1200}$ $= 1.25 \times 10^{-3}\ \Omega$ $R = \frac{\rho l}{A}$ $1.25 \times 10^{-3} = \frac{(1.72 \times 10^{-8})(0.32)}{\pi r^2}$ $r = 1.18 \times 10^{-3}$ m
5	$l = \frac{RA}{\rho}$

	$= \dfrac{(1)[\pi(0.5 \times 10^{-3})^{2}]}{3.5 \times 10^{-5}}$ $= 0.022$ m
6	$R_{eff} = R_1 + R_2 + R_3$ $= 5 + 10 + 15$ $= 30\ \Omega$
7	$\dfrac{1}{R_{eff}} = \dfrac{1}{4} + \dfrac{1}{6}$ $R_{eff} = 2.4\ \Omega$
8	Parallel: $\dfrac{1}{R_{23}} = \dfrac{1}{4} + \dfrac{1}{8}$ $R_{23} = 2.67\ \Omega$ $R_{eff} = R_1 + R_{23} + R_4$ $= 3 + 2.67 + 6$ $= 11.67\ \Omega$
9	$\dfrac{1}{R_{eff}} = \dfrac{1}{100} + \dfrac{1}{R_V}$ $R_{eff} = \dfrac{100R_V}{R_V + 100}$ $V = IR_{eff}$ $2.7 = (30 \times 10^{-3})\left(\dfrac{100R_V}{R_V + 100}\right)$ $R_V = 900\ \Omega$
10	(a) $V_1 = IR_1$ $= (1.11)(10)$ $= 11.1$ V (b) $V_1 + V_{23} = 15$ $11.1 + V_{23} = 15$ $V_{23} = 3.9$ V $V_{23} = V_2 = V_3 = 3.9$ V